EXP 2

```
clc;
clear;
close all;

% Characteristics Study of Side Band Power and Frequencies in AM

% Step 1: Define parameters
Ac = 1;         % Carrier amplitude
Am = 0.5;       % Message signal amplitude
fc = 1000;      % Carrier frequency (Hz)
fm = 100;       % Message frequency (Hz)
fs = 10000;     % Sampling frequency (Hz)
t = 0:1/fs:0.05; % Time vector

% Step 2: Calculate modulation index
m = Am / Ac;

% Step 3: Generate message and carrier signals
message = Am * cos(2*pi*fm*t);
carrier = Ac * cos(2*pi*fc*t);

% Step 4: Generate AM signal
s = Ac * (1 + m*cos(2*pi*fm*t)) .* cos(2*pi*fc*t);

% Step 5: Calculate sideband frequencies
USB = fc + fm;   % Upper sideband frequency
LSB = fc - fm;   % Lower sideband frequency

% Step 6: Calculate powers (assuming 1-ohm load)
```

```
Pc = Ac^2 / 2;          % Carrier power
Pusb = (m^2 * Pc) / 4;  % Upper sideband power
Plsb = (m^2 * Pc) / 4;  % Lower sideband power
Ptotal = Pc + Pusb + Plsb;

% Step 7: Display results
fprintf('Modulation Index = %.2f\n', m);
fprintf('Carrier Frequency = %.2f Hz\n', fc);
fprintf('Lower Sideband Frequency = %.2f Hz\n', LSB);
fprintf('Upper Sideband Frequency = %.2f Hz\n', USB);
fprintf('Carrier Power = %.4f W\n', Pc);
fprintf('Lower Sideband Power = %.4f W\n', Plsb);
fprintf('Upper Sideband Power = %.4f W\n', Pusb);
fprintf('Total AM Power = %.4f W\n', Ptotal);

% Step 8: Plot message signal
figure;
plot(t, message, 'b', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Message Signal');

% Step 9: Plot carrier signal
figure;
plot(t, carrier, 'r', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Carrier Signal');
```

```
% Step 10: Plot AM signal
figure;
plot(t, s, 'k', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Amplitude Modulated (AM) Signal');

% Step 11: Calculate and plot frequency spectrum
N = length(s);
Y = fft(s);
P2 = abs(Y/N);
P1 = P2(1:floor(N/2)+1);
P1(2:end-1) = 2*P1(2:end-1);
f = fs*(0:floor(N/2))/N;

figure;
plot(f, P1, 'LineWidth', 1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('AM Frequency Spectrum');
xlim([0 1500]);

% Step 12: Display conclusion
disp('AM Sideband Analysis Completed Successfully.');

fprintf('Carrier Frequency = %.2f Hz\n', fc);
fprintf('Lower Sideband Frequency = %.2f Hz\n', LSB);
fprintf('Upper Sideband Frequency = %.2f Hz\n', USB);
fprintf('Carrier Power = %.4f W\n', Pc);
```

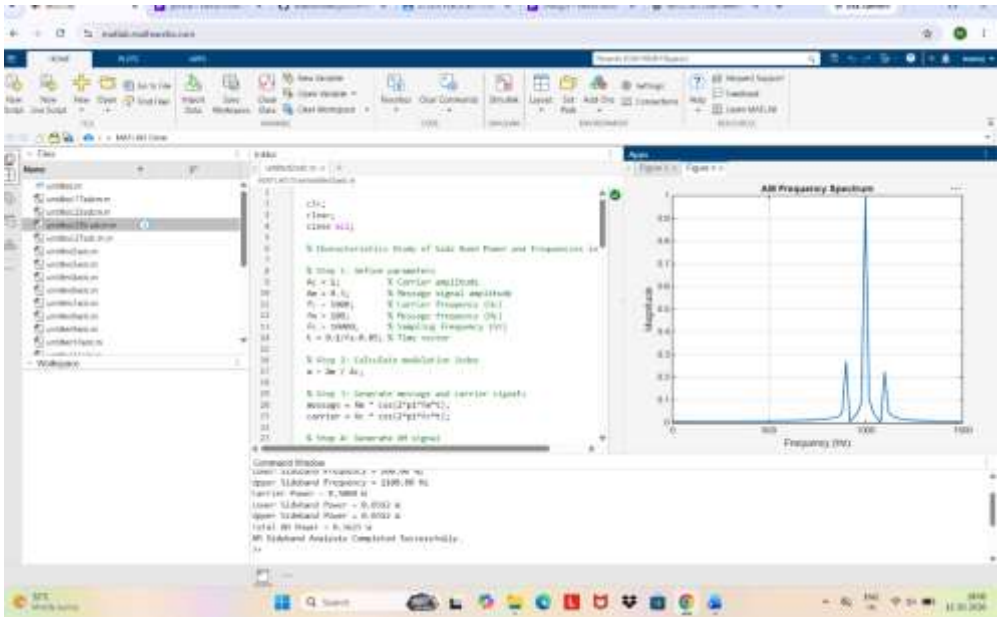
```
fprintf('Lower Sideband Power = %.4f W\n', Plsb);
fprintf('Upper Sideband Power = %.4f W\n', Pusb);
fprintf('Total AM Power = %.4f W\n', Ptotal);

% Step 8: Plot message signal
figure;
plot(t, message, 'b', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Message Signal');

% Step 9: Plot carrier signal
figure;
plot(t, carrier, 'r', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Carrier Signal');

% Step 10: Plot AM signal
figure;
plot(t, s, 'k', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Amplitude Modulated (AM) Signal');

% Step 11: Calculate and plot frequency spectrum
N = length(s);
Y = fft(s);
```


AM Frequency Spectrum
Magnitude
Frequency (Hz)